

Spatiotemporal Flood Risk Modelling Under a Single-Orbit SAR Constraint, Part II: Flood-Aware Emergency Routing and Hybrid Quantum Dispatch

Abstract—Part I of this work derived a calibrated, confound-controlled flood-risk surface $R(\mathbf{x}, t) = S(\mathbf{x}) \cdot T(t)$ for Dakshina Kannada, Karnataka, under a twelve-day single-orbit Sentinel-1 constraint. This paper applies that surface to emergency response over a real 11,910-node, 28,529-edge OpenStreetMap road graph of Mangaluru, of which 17.1% of edges are flood-prone. Flood-aware routing cuts mean route exposure by 33.0%, from 0.171 to 0.114, across the 42 hospital-incident pairs routable under both policies, for a mean detour of 3.8 minutes; the reduction holds at 31.4% over 200 randomly sampled incident sites, so it is not an artefact of the chosen locations. A survivorship-corrected 7×7 sweep of the two cost-function constants separates them: the penalty weight κ is immaterial above ≈ 6 , while the impassability threshold τ governs a step change in connectivity, with 0, 7 and 29 severed pairs at $\tau \geq 0.65$, 0.55 – 0.60 and ≤ 0.50 respectively, so τ is a policy choice about acceptable risk and not a convention. Dijkstra and A* are implemented explicitly to expose their explored sets: across 42 pairs A* returns the identical optimal path every time while settling a median 31.1% fewer nodes, yet is faster in wall-clock on only 25 of them, because heuristic evaluation costs more in interpreted Python than the expansions it avoids. A 9-qubit QUBO/QAOA dispatch layer reproduces the Hungarian optimum exactly and improves with circuit depth over five seeds, with no advantage claimed at this scale. Robustness to the choice of incident sites is tested on 200 randomly sampled nodes.

Index Terms—Flood-aware routing, emergency dispatch, shortest-path algorithms, QUBO, quantum approximate optimization algorithm, sensitivity analysis, road networks.

I. INTRODUCTION

Part I of this study [1] addressed a constraint that shapes every flood-mapping effort over Dakshina Kannada, Karnataka: exactly one Sentinel-1 orbit geometry covers the district, on a nominal twelve-day revisit, so no individual flood can be mapped from SAR because a coastal flash flood rises and recedes inside the gap. That paper responded by decomposing risk into a static, SAR-trained susceptibility surface $S(\mathbf{x})$ and a dynamic, rainfall-driven trigger $T(t)$, fused as $R(\mathbf{x}, t) = S(\mathbf{x}) \cdot T(t)$, and subjected the static term to a three-stage validation ladder that lowered its headline ROC-AUC from 0.9878 to 0.871 as each source of optimism was removed.

A risk surface is not by itself an operational product. The question this paper addresses is what that surface is worth once it has to move an ambulance. We couple $R(\mathbf{x}, t)$ to a real drivable road network for Mangaluru city, obtained from OpenStreetMap, and ask three things of it: how much exposure flood-aware routing actually removes and at what cost in response time; how sensitive that trade is to the two constants in the edge-cost function, which the conference version of this work fixed without justification; and whether the assignment

of vehicles to incidents, the one stage small enough for near-term quantum simulation, can be posed and solved as a QUBO with a verifiable classical reference.

Throughout we keep the reporting discipline of Part I. Where a result runs against the expected direction we report it as measured: A* explores strictly fewer nodes than Dijkstra on every benchmarked pair and is nonetheless slower on a majority of them, and the optimal dispatch assignment does not change between dry and flood conditions even though we expected re-allocation.

A. Contributions of Part II

- 1) A flood-aware routing formulation over a real 28,529-edge network, with exposure and detour measured across all hospital-incident pairs rather than on a single illustrative route.
- 2) A survivorship-corrected sensitivity analysis of the edge-cost constants that distinguishes a genuinely immaterial parameter from one sitting beside a topological discontinuity.
- 3) A search-transparent Dijkstra and A* benchmark that instruments the explored set, reporting a wall-clock result that contradicts the node-count result and explaining the mechanism.
- 4) A hybrid QUBO/QAOA dispatch layer verified against the exact classical optimum and scoped explicitly as a feasibility demonstration.
- 5) A robustness check over 200 randomly sampled incident sites, testing whether the benchmark depends on the choice of the seven named ones.

II. RELATED WORK

A. Emergency routing and dispatch

Risk-weighted shortest-path routing under flooding has an established literature [2, 3]. A* [4] remains the standard goal-directed refinement of Dijkstra [5], with admissibility guaranteeing optimality. Emergency-vehicle siting and allocation is itself a mature operations-research field, from maximal-covering formulations [6] to the ambulance location and relocation models surveyed by Brotcorne et al. [7]; our contribution is not a new formulation but the coupling of that assignment to a SAR-grounded, time-varying risk surface. Ambulance-to-incident assignment is a classical linear assignment problem solved exactly by the Hungarian method [8], which is why any quantum treatment at this scale must be framed as feasibility, never as advantage.

B. Quantum optimization

QAOA [9] approximates combinatorial optima on gate-model hardware; Lucas [10] gives the standard Ising encodings, including assignment. Zhou et al. [11] introduced the INTERP parameter-transfer heuristic used here. Known obstacles, barren plateaus [12] and noise-limited depth, mean that near-term demonstrations at practical scale remain out of reach; our scoping in Section V-D reflects that.

III. STUDY AREA AND NETWORK

The study area, the coastal district of Dakshina Kannada centred on Mangaluru, is described in full in Part I [1], together with the satellite, terrain and climate inputs behind the risk surface used here. This paper adds one dataset: the drivable road network for the Mangaluru routing window, obtained from OpenStreetMap through the Overpass API with `osmnx` [13] and acquired independently of the Earth Engine pipeline. It comprises 11,910 nodes and 28,529 directed edges. Each edge carries an OSM-derived free-flow travel time and is sampled against the susceptibility raster at its midpoint, giving a per-edge $S(e)$. Under the static surface, 4,873 edges, or 17.1% of the network, are flood-prone at $S(e) > 0.5$ (Figure 1).

Seven hospitals and seven incident sites in flood-affected localities give the 49 origin-destination pairs used throughout. The hospitals are real facilities; the incident sites are chosen by us, and Section V-A tests whether that choice matters by resampling them at random.

IV. METHODOLOGY

A. Flood-aware graph and routing

The drivable network is extracted from OpenStreetMap (11,910 nodes, 28,529 edges) and each edge sampled against $S(\mathbf{x})$ at its midpoint. Edge cost under trigger T is

$$c(e) = \begin{cases} t(e) (1 + \kappa R(e)), & R(e) < \tau, \\ \infty, & R(e) \geq \tau, \end{cases} \quad R(e) = S(e) \cdot T, \quad (1)$$

with $t(e)$ free-flow travel time (s), κ a dimensionless penalty weight and τ the impassability threshold. The operating point is $\kappa = 8$, $\tau = 0.60$; Section V-B sweeps both.

Dijkstra [5] and A^* [4] are implemented explicitly, not called from a graph library, specifically so the set of nodes each expands is observable; that set is the object of comparison. The A^* heuristic is a straight-line travel-time lower bound,

$$h(u, g) = \frac{d_{\text{hav}}(u, g)}{v_{\text{max}}}, \quad v_{\text{max}} = 22 \text{ m s}^{-1}, \quad (2)$$

with d_{hav} the haversine distance. Admissibility requires

$$h(u, g) \leq h^*(u, g) \quad \forall u, \quad (3)$$

where h^* is true optimal remaining cost. Since no edge in the graph permits travel faster than v_{max} and the haversine distance lower-bounds path length, (3) holds; we additionally verify it numerically against the graph's own edge weights before reporting any result, and cross-check both implementations against `networkx` as a third reference.

B. Hybrid quantum dispatch

A 28,529-edge routing problem is far beyond exact quantum simulation, so the quantum layer is applied only where the problem is genuinely small: assigning n ambulances to n incidents.

a) *Why $n = 3$ when the rest of the paper uses seven of each:* The one-hot encoding below needs n^2 qubits, so the operational 7×7 problem would require 49. Exact statevector simulation stores 2^{n^2} complex amplitudes: 49 qubits is 5.6×10^{14} amplitudes, roughly 9 PB at double precision, and the brute-force and Hungarian cross-checks that license every correctness claim here would be equally infeasible. At $n = 3$ the state space is 512 amplitudes and every verification is exact. We therefore solve a 3×3 sub-instance formed by taking the three hospitals with ambulance bases in the Mangaluru core (Wenlock, AJ Kuntikana, KMC Attavar) and the three incident sites with the highest mean susceptibility among the seven (Kottara Chowki, Kulur, Pumpwell), that is, the sub-problem an operator would actually face first. The cost matrix C is taken from the same flood-aware routing layer used throughout, so the quantum stage consumes the paper's own risk surface rather than a synthetic instance. The reduction is a constraint of exact simulation and verification, not of the formulation: Equations (4)–(6) are stated for general n .

With binary $x_{ai} = 1$ iff ambulance a serves incident i , and risk-aware travel times C_{ai} from the classical layer, the QUBO is

$$\min_{x \in \{0,1\}^{n^2}} \sum_{a,i} C_{ai} x_{ai} + P \sum_a \left(\sum_i x_{ai} - 1 \right)^2 + P \sum_i \left(\sum_a x_{ai} - 1 \right)^2, \quad (4)$$

with penalty $P = 2 \max_{a,i} C_{ai} + 1$, which is sufficient to make every constraint-violating state higher in energy than any feasible one. Substituting $x_j = (1 - z_j)/2$, $z_j \in \{-1, +1\}$ gives the Ising cost Hamiltonian [10]

$$H_C = \sum_{j < k} J_{jk} \sigma_j^z \sigma_k^z + \sum_j h_j \sigma_j^z + \text{const.} \quad (5)$$

QAOA [9] prepares

$$|\psi_p(\gamma, \beta)\rangle = \prod_{\ell=1}^p e^{-i\beta_\ell H_B} e^{-i\gamma_\ell H_C} |+\rangle^{\otimes n^2}, \quad H_B = \sum_j \sigma_j^x, \quad (6)$$

minimising $\langle \psi_p | H_C | \psi_p \rangle$ over (γ, β) by COBYLA [14], a derivative-free trust-region method appropriate here because the expectation is evaluated exactly but has no cheap analytic gradient. Depth $p + 1$ is warm-started from depth p by the INTERP rule [11], linearly interpolating the optimised schedule onto a finer grid. Quality is reported as the approximation ratio

$$r = \frac{E_{\text{max}} - \langle H_C \rangle}{E_{\text{max}} - E_{\text{min}}} \in [0, 1], \quad (7)$$

with $E_{\text{min}}, E_{\text{max}}$ from exact enumeration of all 2^{n^2} states. For $n = 3$ this is 9 qubits and 512 states, so the ground state is verified by brute force *and* against the Hungarian algorithm [8] on the raw cost matrix. Simulation is exact statevector; no physical quantum hardware is used.

Because COBYLA is a local optimiser on a non-convex landscape, a single run at depth p is one draw from a distribution, not the performance at that depth. We therefore repeat every depth over five random seeds and report mean and standard deviation.

C. Implementation

The pipeline is Python 3.13 with pinned dependencies: `earthengine-api` and `geemap` for server-side processing, `scikit-learn` [15], `xgboost` [16] and `shap` [17] for modelling, `osmnx` [13] and `networkx` for network acquisition and reference validation, `geopandas/rasterio/shapely` for geometry, `numpy` [18] and `scipy` [19] for the statevector simulation and the Hungarian reference solver, and `matplotlib` [20] for all figures. Artifacts are built by a dependency-aware Makefile, so a full rebuild regenerates every number and figure reported here from source data. A suite of 80 automated tests encodes the methodological invariants as executable assertions, including class balance and domain confinement of the sample, $VIF < 10$ for all retained factors, monotonicity of $T(t)$, heuristic admissibility, agreement of Dijkstra, A^* and `networkx` on every benchmarked path, and equality of the QUBO ground state with the Hungarian optimum. Every experiment reported here ran on one Apple M3 Pro workstation with 18GB RAM. Apart from the server-side raster work that Earth Engine performs on its own infrastructure, no additional compute was needed.

V. RESULTS AND DISCUSSION

A. Flood-aware routing

Under the May 2025 trigger ($T = 0.701$), 4873 of 28,529 edges (17.1%) carry $S > 0.5$ (Figure 1). For the representative Wenlock Hospital to Kulur pair, which crosses the Gurupura floodplain, the flood-aware route costs 3.8 min more (13.6 to 17.4 min, +27.8%) and reduces mean exposure by 54% (0.170 to 0.079; Figure 3) (Figure 2). Maximum exposure falls only modestly, 0.593 to 0.509, because both routes must use one of a small number of river crossings. That is a structural property of the network, not a failure of the router, and it is arguably the most operationally relevant single finding here: in this city, risk-aware routing can reduce average exposure substantially but cannot avoid the chokepoints, so resilience investment should target crossing redundancy rather than routing software.

Across all 42 reachable pairs the effect is smaller than the single illustrative pair suggests: mean exposure falls from 0.171 (risk-blind) to 0.114, a 33.0% reduction, for a mean detour of 3.8 min on a mean baseline of 8.9 min over those same pairs. We report the all-pairs figure as the headline because single-pair improvements in this literature are frequently selected favourably.

a) The result is not an artefact of the chosen incident sites: The seven incident locations are author-selected, which conditions the entire benchmark on a choice no reader can audit. Re-running the comparison against 200 incident nodes sampled uniformly at random from the road graph gives an exposure reduction of 31.4% (0.136 to 0.093) over 856

routable pairs, against 33.0% for the curated set: the headline effect is a property of the network and the risk surface, not of the site selection. Two differences are worth noting. The mean detour is much smaller for random sites (1.3 min versus 3.8 min), because randomly drawn nodes are on average closer to a hospital and less likely to sit across a river crossing. And 544 of the 1400 random pairs are severed at this trigger (39%) against 7 of 49 (14%) for the curated set, which says the curated sites are, if anything, *easier* to reach than a typical point in the city.

B. Sensitivity of the routing constants

Figure 4 and Table I report the full 7×7 sweep of (κ, τ) , recomputing all 49 hospital-incident pairs in each of the 49 cells.

a) Correcting for survivorship: A naive reading of this sweep is actively misleading, and the correction matters enough to state before the results. At $\tau \leq 0.50$, 29 of the 49 pairs are severed, so a cell mean taken over "whatever routed" is computed on the 20 survivors, which are systematically the short inland pairs that avoid the river crossings. Compared against a cell at $\tau \geq 0.65$ averaging over all 49, an aggressive threshold appears to *lower* exposure and *shorten* detours when it has really just discarded the hard cases. All statistics below and in Figure 4(a)–(b) are therefore computed over the 20 pairs routable in *every* one of the 49 cells, so the comparison is like with like; panel (c) reports the severed count, which is the quantity that legitimately varies. The risk-blind baseline over that common subset is mean exposure 0.111 over 5.57 min.

b) The penalty weight barely matters above $\kappa \approx 6$: At $\tau = 0.60$ over the common subset, mean exposure changes by less than 1% between $\kappa = 8$ and $\kappa = 16$. Once the penalty is large enough that the router prefers any available lower-risk alternative, increasing it changes nothing, because the residual exposure lies on segments with no alternative. $\kappa = 8$ is defensible not because it is optimal but because the outcome is indifferent to it across a wide neighbourhood.

c) The impassability threshold is a different kind of parameter: Holding $\kappa = 8$, severed pairs go 29 at $\tau \leq 0.50$, 7 at $\tau \in \{0.55, 0.60\}$, and 0 at $\tau \geq 0.65$. That is a step function rather than a gradient, because passability turns on a small number of critical crossing edges whose risk values cluster tightly. The operating point sits one step above a regime in which well over half the service pairs become unreachable.

Once survivorship is removed, the exposure spread across the eight cells neighbouring the operating point falls from 15.9% to 2.4%, so the *exposure* outcome is far more stable than the uncorrected sweep suggested. The consequential variation was never in exposure; it is in connectivity. The conclusion is accordingly not that the parameters are stable, but that one of them is immaterial while the other decides which parts of the city remain reachable at all. τ should be set by policy on acceptable risk, informed by panel (c), and calibrated locally against observed road closures rather than inherited as a modelling convention.

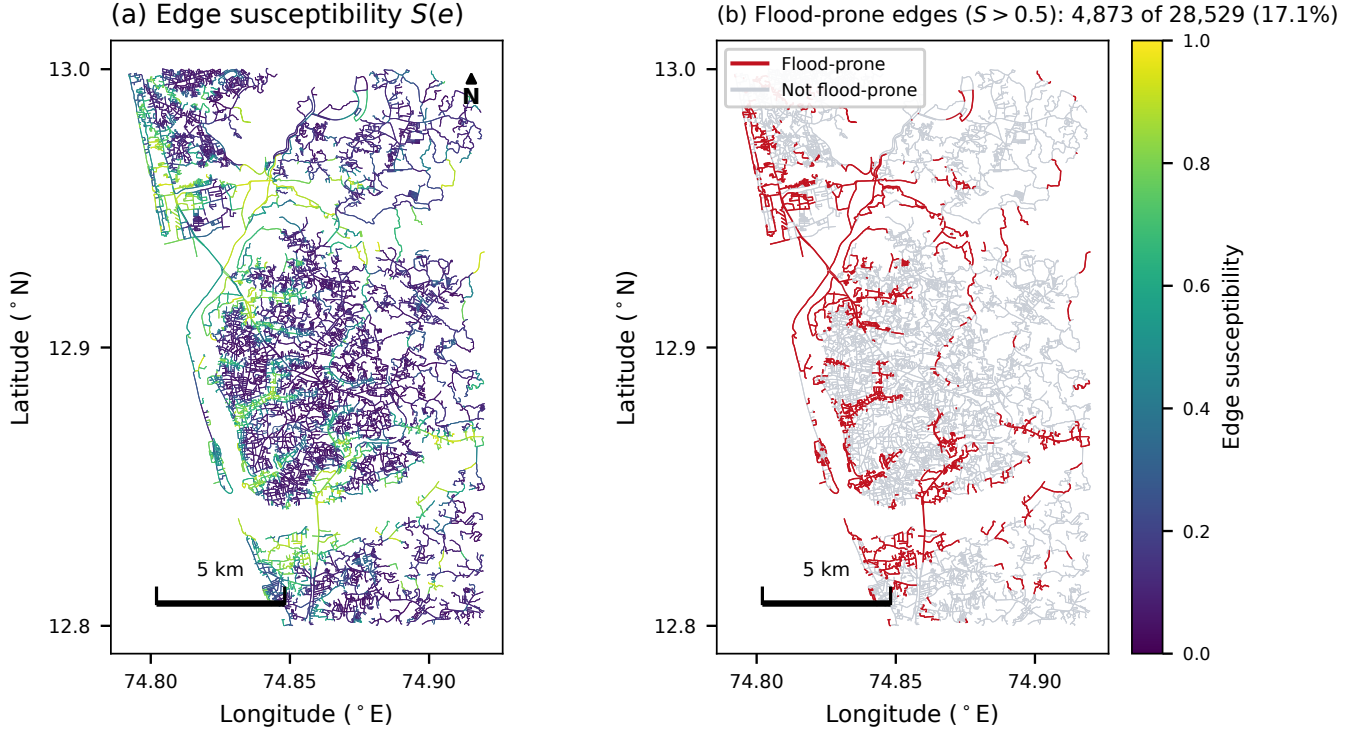


Fig. 1: (a) The routing graph coloured by per-edge susceptibility. (b) The 4873 flood-prone edges ($S > 0.5$), 17.1% of the network, concentrated along the tidal margin and the river corridors. Statistics are computed on the full 28,529-edge graph.

TABLE I: Sensitivity sweep, selected sections. Exposure and detour are computed over the 20 pairs routable in *every* cell of the sweep, so columns are comparable; the severed count is over all 49. Risk-blind baseline on that common subset: mean exposure 0.111 over 5.57 min.

$\tau = 0.60$ fixed			$\kappa = 8$ fixed			
κ	Exposure	Detour (min)	τ	Exposure	Detour (min)	Severed
2	0.0909	0.14	0.45	0.0741	0.56	29
4	0.0871	0.18	0.50	0.0741	0.56	29
6	0.0791	0.32	0.55	0.0787	0.37	7
8	0.0797	0.33	0.60	0.0797	0.33	7
10	0.0788	0.38	0.65	0.0797	0.33	0
12	0.0769	0.45	0.70	0.0797	0.33	0
16	0.0769	0.45	0.75	0.0797	0.33	0

C. Dijkstra versus A*: a multi-query benchmark

Table II and Figure 5 report all 42 reachable hospital-incident pairs under the May 2025 trigger. Seven of the 49 originate from one hospital whose road approaches are entirely severed at this trigger, itself a legitimate finding and not a data problem.

Across the full set of 42 pairs the two searches agreed exactly, matching on both the recovered path and its cost in every case, which is the empirical consequence of the verified admissibility of Equation (2). A* settled strictly fewer nodes on all 42, with a median reduction of 31.1%.

Two nuances only a multi-query benchmark exposes. First, the benefit is highly uneven: the reduction ranges from 52.9% down to 0.03%. The worst case is the Bunder incident on the

tidal margin, where the road network’s connectivity forces a circuitous approach and straight-line distance is a very poor proxy for road distance, making the admissible heuristic nearly uninformative (7991 versus 7989 nodes). Second, and running against the node-count result, A* is faster in wall-clock time on only 25 of 42 pairs, with a higher mean (10.41 ms versus 8.70 ms). Evaluating the haversine heuristic at every expansion costs more in interpreted Python than the expansions it saves. We report this without adjustment: it is a genuine property of a search-transparent reference implementation, and it would reverse in a compiled router, but presenting only the node-count result would misrepresent what was measured.

TABLE II: Dijkstra versus A* across 42 reachable hospital-incident pairs at $T = 0.70$.

Metric	Dijkstra	A*
Mean nodes explored	3349	2933
Median per-pair node reduction	31.1%	
Range of per-pair reduction	0.03% – 52.9%	
Pairs where A* explores fewer	42 / 42	
Mean wall time	8.70 ms	10.41 ms
Pairs where A* is faster	25 / 42	
Identical optimal path and cost	42 / 42	

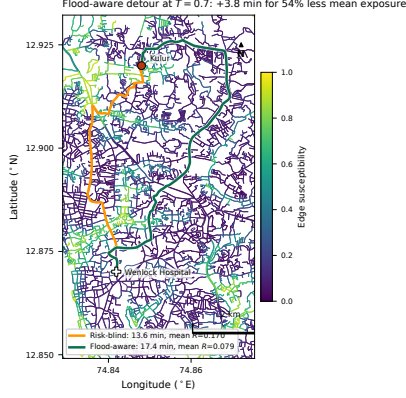


Fig. 2: Risk-blind (orange) versus flood-aware (green) route from Wenlock District Hospital to Kulur at $T = 0.70$, over the susceptibility-coloured network. The flood-aware route takes an inland detour, trading 3.8 min for a 54% reduction in mean exposure.

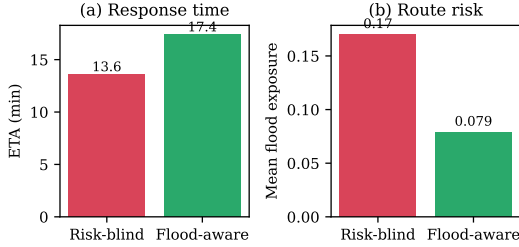


Fig. 3: Risk-blind versus flood-aware routing for the representative pair: response time and mean route exposure.

D. Hybrid quantum dispatch

The QUBO ground state matches the Hungarian optimum exactly in both scenarios: 20.900 min for dry conditions and 22.130 min under the May 2025 trigger. Table III and Figure 6 report the depth study over five seeds.

Averaged over seeds, the approximation ratio improves with depth in both scenarios, from 0.8824 ± 0.0022 to 0.9248 ± 0.0148 (dry) and 0.8695 ± 0.0050 to 0.9209 ± 0.0059 (flood). The improvement is real but saturates: essentially all of it is realised by $p = 3$, and the $p = 4 \rightarrow 5$ increments ($+0.0010$ dry, $+0.0011$ flood) are an order of magnitude smaller than the corresponding seed-to-seed standard deviations, so we do not claim a demonstrated increase beyond $p \approx 3$ –4. Even

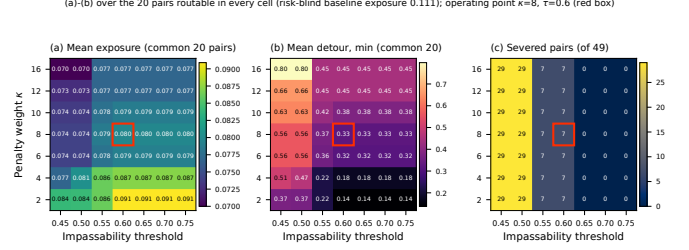


Fig. 4: Sensitivity of the routing outcome to the penalty weight κ and impassability threshold τ at $T = 0.70$. (a) Mean route exposure and (b) mean detour, both over the 20 pairs routable in *every* cell of the sweep so that cells are compared like with like; (c) severed pairs, over all 49. Vertical banding shows the outcome depends on τ far more than on κ ; the red box marks the operating point.

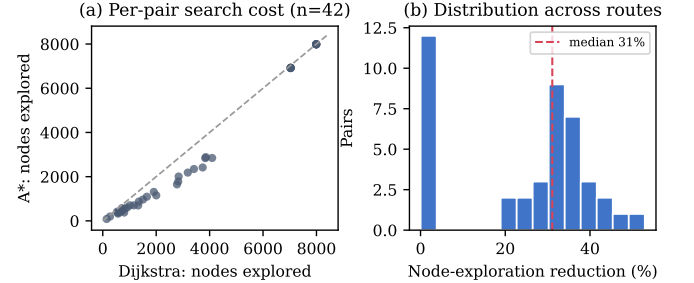


Fig. 5: (a) Per-pair search cost; points below the diagonal indicate A* explored fewer nodes. (b) Distribution of per-pair node-exploration reduction, median 31.1%.

the ordering that does hold is a multi-seed result: individual seeds produce non-monotone sequences, because COBYLA is a local optimiser on a non-convex landscape and the INTERP warm start does not guarantee improvement.

The probability of directly sampling the optimum rises from roughly 2% at $p = 1$ to a peak near 7% at $p = 4$, between $9\times$ and $40\times$ the uniform random baseline of $1/512$, and is plainly *not* monotone: it falls at $p = 5$ in both scenarios. With standard deviations comparable to the means at higher depths, it should be read as a trend with a plateau, not a specification.

The optimal assignment is identical under dry and flood conditions. We had expected re-allocation and did not observe it: at this fleet size and geometry, flooding changes travel times enough to alter route *shape* (Section V-A) but not enough to

alter the assignment's relative ordering. The finding is reported as observed.

We state plainly that at $n = 3$ the Hungarian algorithm is exact and effectively instantaneous, while the QAOA simulation takes seconds and returns the optimum only probabilistically. This layer is a feasibility demonstration of the QUBO/QAOA formulation on a real risk-aware cost matrix, not a claim of quantum advantage. Its value is that the encoding, the constraint penalties and the verification path are in place should problem sizes and hardware make the question live.

E. Limitations

- **The impassability threshold is a heuristic, not hydraulics.** τ marks a susceptibility level, not a fordable water depth, and the sweep shows the routing outcome turns on it.
- **Travel times are free-flow.** No traffic model, no signal delay and no weather effect on speed is included, so absolute response times are optimistic even where the relative comparison holds.
- **The wall-clock comparison reflects an interpreted implementation** chosen for search transparency; a compiled router would convert the node-count advantage into time saved.
- **The quantum layer is a feasibility demonstration** at 3×3 . The Hungarian algorithm is optimal and effectively instantaneous at this size, and no quantum advantage is claimed.
- **Dispatch is single-shot.** Vehicles are assigned once, with no queueing, no re-routing in flight and no fleet dynamics.

F. Comparison with prior work

Risk-weighted routing under flooding is established [2, 3], and QAOA has been applied to vehicle routing and assignment at comparable scale [9–11]. What is different here is the provenance of the weights: the per-edge risk is not an assumed inundation scenario but a SAR-grounded, calibration-audited surface carried over from Part I, and the cost constants that convert it into a route are swept rather than asserted.

VI. CONCLUSIONS

This paper took the risk surface built in Part I into the response layer and measured what it buys.

- 1) **Flood-aware routing removes a third of mean route exposure** across all hospital-incident pairs, for a detour of a few minutes on a representative route. The gain is real but bounded by the network: both the risk-blind and the flood-aware route must cross the same small number of river crossings.
- 2) **The two cost constants are not equally consequential.** The penalty weight is immaterial above $\kappa \approx 6$; the impassability threshold governs a step change in connectivity and must therefore be set by policy on acceptable risk, not by convention. Comparing sweep cells naively invites survivorship bias, because an aggressive threshold silently drops the hardest pairs, so cells are compared on the subset routable everywhere.

- 3) **Search-transparent benchmarking exposes a trade single examples hide.** A* returned the identical optimal path on all 42 pairs and explored a median 31.1% fewer nodes, yet was faster in wall-clock time on only 25 of them, because heuristic evaluation costs more in interpreted Python than the expansions it avoids.
- 4) **The QUBO/QAOA dispatch layer reproduces the Hungarian optimum exactly** and improves with circuit depth over five seeds, at a problem size where it offers no advantage. We report it as a feasibility result and nothing more.

Future work follows from the limitations of both parts. A calibrated depth regression would replace the heuristic impassability threshold with a hydraulic criterion and make τ a physical quantity. A spatially resolved rainfall field would make the trigger genuinely spatiotemporal. A compiled routing implementation would isolate the algorithmic advantage of A* from interpreter overhead, and executing the QAOA circuits on physical hardware would characterise the noise behaviour exact simulation cannot.

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TABLE III: QAOA depth study, 9 qubits, exact statevector, mean $\pm$ s.d. over 5 seeds with INTERP warm starting. “Lift” is mean optimum-sampling probability relative to the uniform baseline $1/2^9$.

p	Scenario	Params	Approx. ratio	Best	P(optimum)	Lift
1	Dry	2	0.8824 ± 0.0022	0.8864	0.0226 ± 0.0026	$11.6\times$
2	Dry	4	0.9116 ± 0.0067	0.9209	0.0503 ± 0.0125	$25.8\times$
3	Dry	6	0.9222 ± 0.0059	0.9301	0.0620 ± 0.0120	$31.7\times$
4	Dry	8	0.9238 ± 0.0074	0.9349	0.0704 ± 0.0263	$36.0\times$
5	Dry	10	0.9248 ± 0.0148	0.9414	0.0531 ± 0.0231	$27.2\times$
1	Flood	2	0.8695 ± 0.0050	0.8738	0.0168 ± 0.0074	$8.6\times$
2	Flood	4	0.8976 ± 0.0107	0.9138	0.0323 ± 0.0163	$16.5\times$
3	Flood	6	0.9179 ± 0.0071	0.9283	0.0529 ± 0.0154	$27.1\times$
4	Flood	8	0.9198 ± 0.0042	0.9268	0.0781 ± 0.0429	$40.0\times$
5	Flood	10	0.9209 ± 0.0059	0.9312	0.0697 ± 0.0391	$35.7\times$

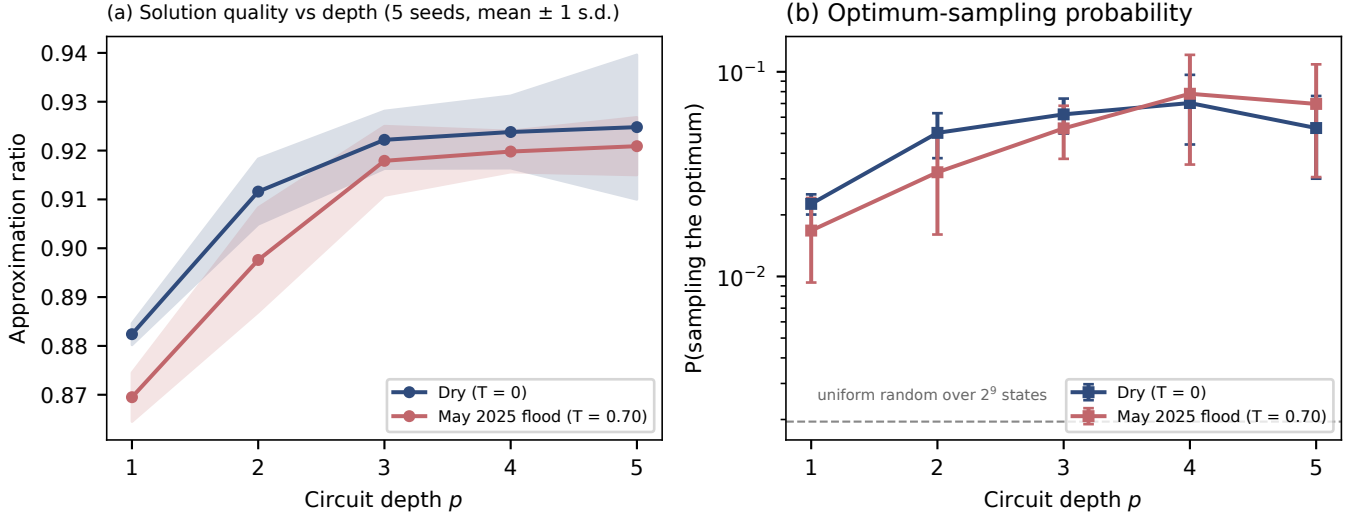


Fig. 6: QAOA depth study over five seeds. (a) Approximation ratio, mean $\pm$ 1 s.d.; the trend is monotone in the mean for both scenarios. (b) Optimum-sampling probability against the uniform baseline over 2^9 states.

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